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Review Article

Neglected topics in chronic pain treatment outcome studies: determination of success

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Summary Although literature on chronic pain treatment outcome has made substantial strides in improving the quality of the studies reported, there remain a number of factors that lead to qualification of the generally positive results. In the two previous papers in this series a set of mitigating factors was discussed, namely, representativeness of the samples treated in these outcome studies, relapse, and non-compliance with therapeutic recommendations. Additional limitations include the lack of agreement on the criteria on which to base evaluation of the success of treatment outcome and the percentage of treated patients included in follow-up data. In this paper, the most common methods for determining success are described (group effects based on standard and quasi-standard outcome measures). The limitations of this approach are discussed and alternative strategies are presented that focus not only on traditional criteria based on group means but on additional criteria including: (a) importance of change (i.e., clinical vs. statistical significance), (b) proportion of patients who improve, (c) cost, (d) efficiency in treatment delivery, (e) and consumer acceptance and satisfaction.

Key words: Chronic pain; Treatment outcome; Research methodology; Determining success

Introduction

In the first two papers in this series (Turk and Rudy 1990, 1991), we discussed a range of factors that influence how the results of pain treatment outcome studies are interpreted. Specifically, we discussed the selective filtering in the referral process, impediments to entering treatment (e.g., denial by third-party payers, patient motivation), drop-out rates, non-adherence, and relapse. In this concluding paper we will take up the issue of typical and alternative strategies for evaluating treatment outcome and illustrate some of the difficulties that can interfere with drawing firm conclusions from existing data in the literature.

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Criteria used to evaluate treatment efficacy

A primary problem in evaluating treatment efficacy in pain research is the definition of outcome criteria. Investigators simply do not agree about how patient improvement should be determined. It will be important to recall this point as the various problems in evaluating treatment success are discussed. The definition of 'treatment success' reflects the agenda and relative values of the different parties evaluating treatment. For example, although a patient may define a substantial reduction in pain severity as treatment success, a third-party payer may be unimpressed with this outcome of treatment and choose instead to focus on whether the patient has returned to gainful employment.

A parallel can be drawn between defining efficacy and the clinical phenomenon of reframing. Patients that continue to focus on pain reduction per se are encouraged to take into account improvements in func-

TABLE I

THE DIVERSITY OF OUTCOME CRITERIA USED TO ESTABLISH TREATMENT SUCCESS IN CHRONIC PAIN TREATMENT

1. Physical functional assessment (e.g., degree of straight leg raising, time to complete an obstacle course)
2. Global ratings by health care providers
3. Use of the health care system (e.g., physician visits, hospitalization, seeking additional treatment)
4. Return to work rates
5. Reduction in pain behaviors
6. Self-reports of activity
7. Self-reports of medication use
8. Self-reports of pain intensity
9. Psychological measures of distress or psychopathology (e.g., depression, anxiety).

tional ability, reduction in medication, social harmony, improved mood, and a plethora of other variables deemed by the pain management staff also to be legitimate efficacy measures, or the success of treatment. Nevertheless, as we will conclude, there is no one correct efficacy measure but rather each must be looked at within its appropriate context.

Criteria of successful treatment

Content differences in outcome criteria. The criteria for judging treatment success vary from study to study and are often unspecified. At other times, qualitative judgments are made by the authors based on ambiguous or loosely defined criteria (e.g., "leading normal lives," Roberts and Reinhardt 1980). Few studies attempt to justify the criteria selected. The number and diversity of criteria that have been used to document success can be seen in Table I. The decision rules regarding success have varied considerably from one study to another. Malec et al. (1981) have criticized this lack of standardized outcome criteria stating:

> Poor delineation of criteria for improvement, as well as lack of agreement among studies on variables indicating improvement, makes comparison of results among studies impossible. A clearly defined set of criteria which would indicate clinically significant general improvement for the individual program participants is absent in the follow-up literature (p. 369).

The question remains then how to determine the relative efficacy of various treatments when diverse measures and criteria are used and frequently combined to demonstrate treatment success.

Patient vs. clinician ratings of outcome. Some studies have used patients' self-ratings (Gentry et al. 1977; McCreary et al. 1979; Cassisi et al. 1989) whereas others used physician ratings (Wolff 1971; Wiltse and Rocchio 1975; Jamison et al. 1976; Waring et al. 1976) or a combination of physician and patient estimates to establish the level of treatment success. Daniel et al. (1983) directly contrasted agreement between therapists' and chronic pain patients' perceptions of treatment outcome and found that therapists rated patients as much more improved than did patients themselves.

Vague vs. specific definitions of outcome. The criteria used to establish treatment success vary from vague to quite specific. An example of relatively vague criteria is found in Parris et al. (1987) who included the following as program goals: (a) increase of daily activity levels; (b) reduction of stress; (c) decreased nociception and change of the patient's perception of pain; and (d) return to as normal a lifestyle as possible. How are 'reduction in stress' and 'return to as normal a life style as possible' quantified? How do the authors determine that there has been a reduction in nociception? Who makes these subjective assessments — the patient, the clinical staff, or third-party payers?

Investigators studying outcome in pain management strive to make their studies internally reliable so that changes in the dependent or outcome variables can be attributed to manipulations of the independent variables (the treatments). However, even objective variables used to measure success rates are strongly influenced by local situational factors that are outside the control of the treatment program. For example, employment depends on the local job market, the educational and skill levels of the patients, and the willingness of employers to hire persons with a history of chronic pain problems or compensation. Sheikh (1987) reported that one of the few variables that predicted return to work following a back pain rehabilitation program was general unemployment in the home area. Similarly, Polatin et al. (1989) demonstrated that one of the best predictors of success, defined as return to work, was job availability. Compensation status also is strongly influenced by the local legal structure of compensation and other financial considerations, and medical usage is influenced by local medical practice and insurance coverage (Gallon 1989).

Liberal vs. stringent definitions of successful outcome. Criteria of success vary from liberal to quite stringent. For example, Malec et al. (1981) used the following fairly stringent set of criteria to determine success: (a) reported no use of narcotic analgesics, muscle relaxants, or tranquilizers, (b) employed, in training, or running a household, or continuation of 50–100% of exercises and reported increased recreational activity; and (c) no reported increase in pain. Of the patients providing adequate data, 37% (total $n = 40$) met all 3 criteria. Fredrickson et al. (1988) and Mayer et al. (1985, 1987) also included physical performance measures (e.g., obstacle course, physical measures of strength, endurance, and flexibility) in evaluating treatment success. The treating staff then subjectively rated patients on 5-point scales indicating percent improvement in the specific areas measured and progress toward goals identified at admission, along with patient reports of activity.

Roberts and Rinehardt (1980; see also Guck et al. 1985) used even more stringent criteria to judge success: (a) patients had to be employed and if not employed they had to be either unemployed for reasons other than pain, homemaker by profession, retired for reasons other than pain, or attending school or training; (b) up and active at least 8 h/day; (c) receiving no compensation for pain problems whether employed or not; (d) no pain-related hospitalizations for surgery since treatment; and (e) taking no prescription narcotics or psychotropic medications. Notice that these criteria do not include subjective reports of pain or mood in the outcome criteria because these were not goals of these behavioral programs. These authors argue for the reliance on objective measures of success because of the potential biases in subjective reporting.

Problems with the reliability of outcome measures

All outcome data are collected with some degree of uncertainty and inaccuracy in that no measure of treatment improvement is completely free of measurement error. Thus, the reliability (i.e., replicability, reproducibility, and consistency) of outcome measures is a matter of degree. One of the primary reasons for establishing the reliability of outcome measures is the important influence that reliability can have on the conclusions drawn from treatment studies. Measures that demonstrate good reliability are at least as important a component of a well-designed study as are randomization, double blinding, controlling, when necessary, for prognostic factors, and so forth.

The use of unreliable measures can have numerous untoward consequences including the need to use larger sample sizes to obtain adequate statistical power, statistical analyses that result in biased estimates (e.g., correlations that are greatly reduced in size), and even the selection of biased clinical samples (see Fleiss 1986). In *all* instances, it must be assumed that an outcome measure is unreliable until proven otherwise. This is particularly true for the ratings of success made by health care providers. For these ratings, it is essential to collect ratings from multiple clinicians for each patient and to test the reliability of these judgments before using them to measure treatment improvement. Ideally, the ratings should be made by clinicians who were not directly involved with the treatment of the patient, and they should be blind to the treatment condition.

Methodological problems

In addition to criterion, statistical, and reliability problems, most pain outcome studies also suffer from one or more serious methodological or experimental design problems. For example, despite reported concerns in the literature, (e.g., Turner and Chapman 1982; Aronoff et al. 1983), few studies have utilized waiting-list or placebo control groups. The reasoning implied is that because the patient samples used have long histories of pain, it is not likely that the short interval of treatment would, in and of itself, be sufficient to account for the obtained results (Fordyce 1988). Nevertheless, receiving attention from a therapist or treatment team is likely to make any treatment that is compared to no-treatment condition appear to be effective, at least in terms of patients' self-reports due to demand characteristics. That is, some patients are likely to fulfill their 'expected' role of complaining of symptoms upon arrival and dismissing symptoms upon termination of treatment.

Even when control groups are included, a special set of methodological problems is encountered when attempting to compare the relative efficacy of one treatment modality to another. Are each of the treatments offered equally credible to patients? Differential treatment credibility can influence outcome, independent of the active ingredient of the treatment. If patients do not believe that the treatment is likely to be effective, they will be less likely to adhere to the self-care recommendations (e.g., regular physical exercise, practice of relaxation exercises). Failure to establish the comparability of perceived treatment credibility might result in erroneous conclusions regarding the relative efficacy of one treatment modality compared to another. The credibility of each treatment should be evaluated following presentation of the treatment rationale, at some point during treatment, and at the termination of treatment. Treatment comparison studies should demonstrate that the modalities being compared were equally credible and acceptable to patients. Procedures for assessment of treatment credibility are readily available (e.g., Borkovec and Nau 1972).

Treatment fidelity

Treatment fidelity refers to whether the treatment is conducted in a consistent and specified manner. Assessing fidelity allows one to know exactly what was and was not done with the patients. Without this understanding, it is difficult to compare the results of different studies. For example, Cassisi et al. (1989) summarized outcome studies on multidimensional rehabilitation programs for chronic low back pain and noted that the "reported success rates for rehabilitation programs are very high" (p. 877). At face value, one assumes that we know that these programs are effective and that we should have confidence referring our patients to such a program. But what do we know? Do we know what exactly transpired in the program? Do we know what aspects were important? Cassisi et al. also noted that "these programs vary greatly in terms of staffing, treatment modalities offered and availability of inpatient and/or outpatient facilities" (p. 877). Unless we know what actually occurred in the programs and ensure that the programs are indeed comparable, we cannot justify aggregating results or making global statements. Two treatments specified, for example, as being cognitive-behaviorally or operantly oriented can differ markedly from each other because there is no general consensus about what comprises the ingredients of these treatments. When treatment fidelity is documented, there is a greater likelihood of replication if treatment is provided in the manner specified in the experimental study.

There are several critical features that are required to ensure treatment fidelity. The components of the treatment must be precisely defined. This is most effectively done through the use of treatment manuals specifying the content as well as the process of therapy. Treatment manuals are being utilized with increasing frequency in therapies as diverse as the behavioral treatment of depression (e.g., Lewinsohn et al. 1984), psychoanalytic psychotherapy (e.g., Luborsky 1984), and headache pain management (e.g., Penzien et al. 1989).

It is also essential to adequately train the treating staff and then to assess their initial competence to implement the intended treatment. In order to prevent clinicians from 'drifting away' from the protocol, ongoing supervision of those delivering the treatment is required. Finally, in order to assess and document treatment fidelity, actual samples of treatment should be monitored and independently evaluated through direct or taped sampling.

In order to compare data from the outcome literature with chronic pain programs, we evaluated all treatment outcome studies published during the years 1988 and 1989 identified in a meta-analytic review of the efficacy of multidisciplinary pain treatment centers (Flor et al. 1992). These studies were reviewed for the presence of a treatment manual, explicit statements regarding the use of supervision, and statements regarding the sampling of treatment to determine adherence to protocol. In addition to these measures, the presence of any information regarding personnel involved in treatment or the content of treatment was noted.

The results of this review are illuminating but distressing. Of the 13 treatment outcome studies reviewed, not one mentioned using a treatment manual, none mentioned actual sampling of treatment to assess content and quality, and only one (Simmons et al. 1988) mentioned training and supervision of treatment providers. Although all 13 studies mentioned the types of personnel involved or the contents of the program, these descriptions often were vague and did not allow for detailed understanding of the treatment components included. For example, both Salzberg et al. (1988) and Harkapaa et al. (1989) describe their programs as being based on the "back school" model. It is impossible to determine from the use of the phrase 'back school' what comprised the primary treatment components or if an impartial observer would perceive any similarity between the 2 programs.

The overall conclusion is that we do not actually know what treatments are being delivered in different programs. When studies fail to demonstrate a significant improvement in patients, we do not know if this was caused by inadequate treatment implementation, the inclusion or exclusion of treatment elements, or various other factors that can affect outcome but escape detection without some check on treatment fidelity. Conversely, when results are positive, it is impossible to determine what the essential components of the treatment were or how the treatment was conducted. Thus, generalizing the treatment regimen to other settings is hazardous. On a more positive note, other areas of pain research, most notably studies on the measurement of pain behaviors (e.g., Keefe and Williams 1992), have had a much greater emphasis on the measurement and assurance of fidelity. It will be essential to incorporate this emphasis into current research on treatment outcome.

Reactivity of outcome criteria

In a recent meta-analysis, Flor et al. (1992) assigned dependent measures to 3 classes depending upon reactivity (e.g., potential for patient and/or therapist bias), namely, (1) all physiological measures, laboratory data, blinded rating or behavioral assessments, or medical record data; (2) standardized measures with little connection with the treatment or therapist (activity diaries); and (3) patient self-report of improvement, therapist improvement ratings, or instruments with obvious relationship to treatment outcome. On the basis of these outcome qualitative ratings, only 3% of 65 pain clinic treatment outcome studies were judged to have included measures of low reactivity (class 1). Additionally, only 29% of the studies were rated as using good quality outcome measures, that is, measures with established and acceptable levels of reliability and validity.

It is important to note that the criteria considered above refer to the effects of treatment on patients. Nevertheless, as discussed below, patient change is not the only criterion on which treatment evaluation can or should be based (Kazdin and Wilson 1978). Other broader conceptual and empirical issues pertaining to criteria for evaluating treatment exist that have yet to be addressed adequately in chronic pain research.

These issues embrace the manner in which patient change is evaluated, independent of the specific assessment devices used, the varied effects that treatment may exert on different patients, and the implications of these effects for evaluating different therapy techniques. Additional criteria pertaining to the efficiency, cost, and consumer evaluation should be considered along with traditional measures of patient improvement.

Criteria of success are influenced by the percentage of patients available at follow-up

The issue of the percentage of treated patients available at follow-up when determining relapse was raised in the preceding paper in this series (Turk and Rudy 1991). This issue is of concern when attempting to evaluate the success of various pain rehabilitation interventions. It is not unusual for follow-up evaluations to be based on less than 30% of the originally treated population. For example, in a study by Swanson et al. (1979), only 74 of 200 patients participated in the follow-up phase, although the authors reported that 65% of those treated remained improved 1 year later. This percentage, however, represents only 25% of the original patient sample. In another study, Beekman and Axtell (1985) based their optimistic conclusions regarding the efficacy of their 4-week inpatient program on only 20% of the program completers who were available at time of the 6-month follow-up.

Thus, when determining treatment success the rates reported will depend on whether investigators consider the entire sample referred for treatment, only those patients who were approved for treatment by insurance companies or who chose to accept treatment, whether patients who drop out of treatment are removed from the analyses or are considered as treatment failures, and so forth. An example of what might be considered an 'overly optimistic' approach (see Turk and Rudy 1990) is provided by Beckman and Axtell (1985) who chose to exclude both drop-outs and those unavailable at follow-up before computing treatment outcome statistics. If a 'worst case' scenario is used, as described in the preceding paper in this series (Turk and Rudy 1991), the non-respondents would be treated as treatment failures, that is, without evidence to the contrary it would be assumed that non-respondents would disagree that the treatment program had been beneficial. This approach would lead to very different conclusions about treatment success. For example, Wang et al. (1980) indicated that of the 50% of patients who responded to follow-up, 48% agreed that the treatment program had been beneficial. If the non-responders were treated as treatment failures then the success rate would be reduced to a mere 24%.

It could be argued, of course, that the worst case approach is too stringent since at least some of the non-respondents may have died, relocated, or refused to reply because of reasons other than dissatisfaction. However, without additional information regarding nonrespondents the best conclusion that can be reached is that 'somewhere' between 1/4 and 3/4 of the treated patients in the Wang et al. (1980) study were satisfied with the outcome of the treatment.

Although, of course, treatment efficacy cannot be clearly evaluated for patients who do not participate in the follow-up phase of a study, the relatively small percentage of patients available at the time of follow-up reported in most outcome studies raises serious questions about the generalizability of the results reported (Turk and Rudy 1991). Additionally, it should be noted that when a substantial number of patients are unavailable for follow-up, the follow-up sample cannot be considered a random sample of those treated, which, therefore, violates the basic assumptions of traditional statistical procedures used to determine treatment success.

Some tentative solutions

Many reviewers have called for the use of standardized measures normed on pain patients in treatment outcome studies. Chapman et al. (1985), Turk and Kerns (1983), Turk and Rudy (1987), and Karoly and Jensen (1987), among others, have provided extensive reviews and discussions of many of the conceptual and psychometric issues involved in pain assessment. Thus, we will not review these topics here. However, several recent innovations that may help resolve some of the problem of evaluating treatment outcome within studies as well as between studies are worthy of particular attention.

Use of component or composite scores. A major dilemma in trying to establish treatment outcome within a study is how to treat multiple conceptual and statistically related measures. For example, if a treatment outcome study includes the McGill Pain Questionnaire (Melzack 1975), pain diaries, the Beck Depression Inventory (BDI) (Beck et al. 1961), the MMPI, analgesic medication use, use of the health care system, pain behaviors (Keefe and Block 1982), activity levels, and percentage of patients working, what criteria should be used to establish success? If for example, following treatment patients report being significantly more active and less depressed but report no clinically significant change in self-report of pain (Newman et al. 1978), is the treatment successful? How should we deal with significant reductions in pain behaviors, increased performance of exercise and physical conditioning activities, and self-report of pain but no changes in daily activity levels as reported by Beekman and Axtell (1985)? Finally, if the treatment leads to significant reductions on all measures with the exception of return to work, is the treatment successful?

One empirical strategy to deal with multiple outcome criteria suggested by Kerns et al. (1986) is to compute component scores among statistically related measures derived from principal component analyses and determine the treatment efficacy based on patient changes on component scores. This strategy has the advantage of aggregating measures that are correlated and, therefore, prevents performing numerous statistical analyses that inflate the experimentwise type-I error rate. Additionally, the resulting component scores will most likely be more reliable than the individual scales that comprise the component. A potentially serious limitation with this approach, however, is that the sample sizes required to obtain stable component patterns (see Guadagnoli and Velicer 1988) are larger than those frequently used in pain outcome studies. Moreover, the use of aggregate scores creating several categories of outcome measures still leaves open the possibility of differential outcomes on the various components.

Use of meta-analysis. Comparing outcomes across studies provides a number of difficulties. As noted, a particular problem concerns how to compare studies with diverse outcome measures. The typical review of pain treatment effects has relied on the subjective 'box score' approach in which the reviewer's interpretation of relevant published studies is based on his or her subjective preferences (e.g., Turner and Romano 1984; Chapman 1986; Keefe et al. 1986; Linton 1986; Jessup 1989). In pursuit of more objectivity in the review process, a group of techniques known collectively as 'meta-analysis' was developed that provides a strategy to quantitatively review a range of studies (Glass 1976). This approach has begun to receive attention in evaluating the efficacy of different interventions for pain (e.g., Blanchard et al. 1980; Fernandez and Turk 1989; Flor et al. 1992; Holroyd and Penzien 1986; Malone and Strube 1988; Patel et al. 1989).

Meta-analyses consist of the integration of research findings through statistical analysis of individual studies. Basically, individual effect sizes can be averaged across studies to provide a global portrait of treatment effectiveness, or they can be examined further as a dependent variable moderated by the type of treatment in question. Recently, Fernandez and Turk (1989) demonstrated how quite different, if not opposite, conclusions could be drawn based on reviewing the same-studies examining the efficacy of different pain coping strategies by using qualitative versus quantitative (meta-analytic) methods.

Despite improvements upon previous research review techniques, meta-analyses have not been exempt from criticism (e.g., Strube and Hartman 1983). These criticisms have included inappropriately aggregating data from studies that use very different interventions to treat dissimilar problems, assuming that results obtained with different types of outcome measures and research methodologies provide equally useful information, and for not considering the quality of the studies that are combined. Of course reviews based on qualitative aggregations are likewise not exempt from these criticisms (Flor and Turk 1989). Extended discussion of these and other criticisms of meta-analytic procedures is beyond the scope of the present paper.

Using meta-analytic techniques to evaluate 65 treatment outcome studies, Flor et al. (1992) concluded that pain clinics produce significant improvements in patients on self-report, physical, and behavioral measures that were maintained by the time of follow-up evaluations, some of which ranged up to 3 years (see also Malone and Strube 1988).

It is important to note, however, that meta-analytic pain treatment outcome papers reported outcomes with considerable variability. For example, in the review by Blanchard et al. (1980), outcomes with relaxation training ranged from 14 to 100% reduction in tension headache activity. Such large differences in outcome with supposedly the same intervention strategy suggest that some unspecified variables (whether patient, treatment, or research design) can have a significant impact on treatment outcome.

Alternative criteria of success

Typically, conclusions about treatment or the relative effects of different treatments are drawn on the basis of statistical probability of differences between mean scores of pre–post treatment and/or mean scores of patients receiving various treatments or control conditions. Several other criteria need to be considered, including (a) the importance of change, (b) the proportion of treated individuals who change, (c) the breadth of change, (d) the efficiency and cost of treatments, and (e) consumer evaluation of treatment (Kazdin and Wilson 1978; Hugdal and Ost 1981; Jacobson et al. 1984).

The durability of the treatment effects, broadly defined, was discussed at length in the second paper in this series (Turk and Rudy 1991) and thus we will not examine this topic in this paper. In addition to evaluating criteria designed to measure therapeutic change, we also will consider the efficiency and cost of administering treatment, the financial and emotional costs to patients, and how cost considerations relate to overall outcome. Finally, the acceptability of treatment to patients also will be addressed. In sum, our emphasis will be that multiple criteria are needed to evaluate treatment effectiveness and the relative value of different treatment approaches. The focus on narrowly circumscribed measures of average patient performance obscures the potential value of different treatments in relation to specific therapeutic and social goals.

Significance of change

The significance or importance of change appears to be a simple question until we consider the importance of change *to whom*. What constitutes a successful treatment outcome will depend on whom is asked — the patient, professional staff, or society, including third-party payers (Strupp and Hadley 1977).

Patients enter pain rehabilitation with a number of presenting problems in addition to pain, for example, limited activity, dysphoric mood, excessive use of analgesic medication, and work disability. Different individuals or organizations involved in a chronic pain case will define 'the' most important presenting problem according to their values. Patients often disagree with health care providers and the third-party payer as to the most important presenting problem.

Colvin et al. (1980) noted that the majority of patients indicated that they desired total and permanent relief of pain as their primary goal. Specifically, 97% of the patients (N = 287) indicated that they would accept a 50% improvement (see also Keefe et al. 1986), and 52% indicated that they would try elsewhere if the treatment did not bring acceptable pain relief. At an outpatient rheumatology back pain clinic, Fitzpatrick et al. (1987) confirmed these findings and noted there was a significant inverse correlation between reductions in pain severity and satisfaction with the treatment received. Fully 1/3 of patients interviewed in their homes 3 months following treatment rated their treatments as unhelpful (note that this was based on a response rate of only 55% of treated patients, using the worst case approach described earlier would make the outcome of this study much worse).

The criterion of 50% symptom reduction may be viewed as one benchmark against which to evaluate patients' appraisals of clinical efficacy. Using this criterion, how successful are pain rehabilitation outcome studies? Duckro et al. (1985, 1986) and Newman et al. (1978) report significant improvements on such measures as reduced use of medical resources and prescribed analgesics, improvement in measures of physical functioning, and increased ability to cope with pain. At the same time, patients reported *no* significant change in pain intensity and so these treatments would be rated by patients as failures, at least as far as pain reduction was concerned. Smith (1988) indicated that 49% of treated patients reported that their pain was actually *worse* following the treatment!

We can consider other illustrative studies. Wang et al. (1980) conducted a 1 to 3-year follow-up study of patients treated in an outpatient pain clinic. Of an initial sample of 725 patients treated, 361 (50% return rate) responded to a questionnaire that asked them to indicate their belief as to whether the treatment had been beneficial to them. Examination of pain severity reported by patients at follow-up reveals that 31% indicated that their pain continued to be very severe or unbearable, with an additional 21% indicating that their pain still was fairly severe. Because reduction of pain is the primary reason patients give for seeking treatment, it is possible that the 52% with fairly severe to unbearable pain comprise the proportion of patients who indicated they did not view the treatment program as beneficial. The authors of this paper did not report on the correlation between pain severity and perceived benefits so this speculation cannot be verified.

As an anchor by which to evaluate the average percentage of reduction in patients' reports of pain intensity that can be expected to result from treatment, Chaplin (1991) reviewed a number of treatment outcome studies and concluded that although these studies demonstrated a statistically significant decrease in self-rated pain intensity, the magnitude of reduction tended to be small, ranging from 10 to 20%. In their meta-analytic study, Flor et al. (1992) found that the average reduction in pain intensity for pain clinic patients across the 65 studies reviewed was 37%.

Carron et al. (1985) reviewed studies of patients treated at pain rehabilitation programs in both New Zealand and the United States and noted that minimal reduction in pain was reported at the 1-year follow-up. Interestingly, 39% and 45% of the United States and New Zealand sample, respectively, indicated that they considered themselves at least 'somewhat' improved at the follow-up. Cassisi et al. (1989) reported, 61% of patients who had completed the treatment program reported a modest but statistically significant 27% decrease in pain. They noted that despite the relatively modest reduction in pain, 79% of the group reported that they were satisfied with the way they were treated in the program and 74% said that they would recommend the program to a friend who had a similar condition. These data suggest that patients may use criteria other than pain reduction alone to evaluate the treatments received.

Cassisi et al. (1989) found that patients who completed treatment reported significantly fewer physician visits than those who were not approved for treatment by insurance carriers and that treated patients had significantly fewer surgeries following treatment than patients who declined to participate in treatment. They also reported that 69% of program completers had returned to work by follow-up compared to 41% and 39% return to work for those patients not approved for treatment by insurance carriers and patients who declined treatment on their own, respectively. These percentages can readily be converted to dollars saved, the ultimate criterion of success to insurance carriers (Simmons et al. 1988; Stieg and Turk, 1988; Cicala and

Wright 1989). Despite patients' understandable preoccupation with pain severity, third-party payers may care little about reduction in pain intensity unless these reductions are associated with reduction in health care utilization, reduction in disability payments, and return to work. Thus, it is important that clinical investigators include these types of measures in determining the success of treatment programs.

Clinical versus statistical significance of chances following treatment

Related to the importance of change is the consideration of the clinical versus the statistical significance of treatment outcome. As noted in the previous section, a number of studies report statistically significant reduction in pain intensity that may not be 'clinically significant'.

When considering treatment efficacy, investigators might wish to view the magnitude of improvement not only compared to pre-treatment levels and other chronic patients receiving the same or alternative treatments but also to healthy individuals (Jacobson et al. 1984). How similar are patients to 'normal' levels of functioning following treatment? In one study, Rudy et al. (1991) demonstrated that at discharge from a pain rehabilitation program, patients approached comparable levels of functional capacity as non-pain subjects. These results illustrate a 'normalization' of physical function rather than relying exclusively on within group improvements, that is, patients compared to themselves. Thus, this approach to evaluating treatment success does not categorize patients as 'much improved', 'very much improved', and so on, which commonly has been done in the literature (e.g., Swanson et al. 1979). Rather, the extent to which treatment restores adequate or acceptable levels of functioning can be assessed directly.

Of course, it is important to acknowledge that adopting a normative standard may, at times, be excessively stringent and, therefore, inappropriate. Physical factors associated with an injury may provide an upper limit on the extent of an individual's physical functioning and may prohibit the patient from returning to 'normal' physical functioning. In many ways, evaluating treatment in light of the importance of the therapeutic changes seems to be a more appropriate criterion than is the statistical comparison of group differences.

One strategy that can be adopted in evaluating the importance of treatment is to compute an 'effect size' for each outcome variable. This permits the investigator to determine how much the individual has improved compared, for example, to a control group. The change score could then be divided by the control group standard deviation. This approach is somewhat limited, however, because it does not lead to a decision rule for categorizing people as 'improved' (Jacobson et al. 1984). The investigator also can compute omega-squared to consider the percent of variance accounted for by the treatment (see Bradley et al. 1987).

A potentially valuable alternative is to compute a reliability of change (RC) index proposed by Jacobson et al. (1984) and later amended by Christensen and Mendoza (1986). This index can be calculated as follows:

$$RC = \frac{x_2 - x_1}{S_{diff}}$$

where x_1 represents a patient's pretreatment score, x_2 represents the same patient's post-treatment score, and S_{diff} is the standard error of the difference between the two scores, which provides a measure of the spread of the distribution of change scores that would be expected if no actual change in scores occurs. S_{diff} can be computed from a measure's standard error of measurement (S_E) by the formula:

$$S_{diff} = \sqrt{2(S_E)^2},$$

with S_E defined as

$$S_E = s_1\sqrt{1 - r_{xx}},$$

where s_1 is the standard deviation of a pretreatment experimental group or a control group, and r_{xx} is the test-retest reliability of the measure being used to determine treatment change. Values of RC that exceed 1.96 are unlikely to occur ($P < 0.05$) unless an actual change in scores occurs between pretreatment and post-treatment assessment. That is, changes that exceed this magnitude can be considered as reflecting more than the normal measurement fluctuations that occur with repeated testing with a measure that has less than perfect reliability.

Recently, Harkapaa et al. (1989) computed RC indices to compare the effectiveness of inpatient and outpatient treatment approaches for back pain. They created a low back pain disability index that included 15 items describing disability caused by back pain during the past month. Scores on this scale ranged from 0 to 45 with an internal consistency value (coefficient alpha) of 0.89. RC calculations at an alpha value of 0.05 indicated that at least a 5-point change on the disability scale was necessary to suggest clinically significant improvement had occurred between pre-treatment and follow-up assessments. At a 2.5-year follow-up, the authors reported that 25% of the treated inpatients exceeded the critical cutoff criteria on the disability scale. This improvement rate can be compared to the 14% and 10% improvement rates (also determined by the RC index) observed for patients treated on an outpatient basis and a control group of low back patients, respectively. Chi-square tests of these improvement rates indicated that (1) the inpatient and control groups were statistically different, (2) there was a 'marginally' significant difference favoring the inpatient over the outpatient groups, and (3) no significant difference was found between the outpatient and control groups. Thus, although statistically the inpatient treatment was most effective, the clinical significance of a 25% improvement rate in patients receiving inpatient treatment is open to alternate interpretations.

The Harkapaa et al. (1990) findings as well as the basic logic of the RC index are not without some rather substantial limitations. First, Harkapaa et al. (1989) did not randomly assign patients to the three experimental conditions, which seriously limits the conclusions that can be reached from their findings. Second, an equally serious limitation that impacts on computing RC indices, in general, is the implicit assumption of patient homogeneity. The statistical implications of this assumption can best be illustrated by considering the formula for computing S_E values, described above. This formula requires an estimate of the standard deviation of a measure that is derived from group data, based on the pretreatment assessment. The magnitude of this estimate will increase as the patients comprising the group under consideration become increasingly heterogeneous. As a consequence, the value of S_{diff} will likewise increase and because this value is used in the denominator of the RC formula, the magnitude of the difference between pretreatment and post-treatment scores (x_2–x_1) that is needed to reach the 0.05 cutoff level will similarly increase. Thus, for example, identical treatments provided to patient groups with differing levels of heterogeneity will produce differing 'success rates' from a RC perspective, even though the magnitude of the difference scores are identical among the patient groups. In sum, computing RC indices should be approached with caution and, at least for many of the measures used to evaluate treatment success (e.g., pain severity, depression, disability), the investigator should not assume that pain patients form a homogeneous group (Turk and Rudy 1988, 1990).

A third potential problem with computing RC indices as well as more traditional group-oriented inferential statistics (repeated measures) relates to the measures that are selected to quantify change. Although there are a large number of pain assessment instruments and procedures that have been developed to measure outcome, some of which have been subjected to extensive psychometric testing to establish their reliability and validity, the sensitivity of these measures to detecting change due to treatment effects is rarely reported. For example, Deyo and Inui (1984) examined the sensitivity of the widely used Sickness Impact Profile (Bergner et al. 1981) and concluded that it did not contain sufficient precision to measure clinical improvement in individual patients. Thus, if investigators select measures that are not sensitive to treatment effects, conclusions about treatment improvements may be biased by the inability of the measures to detect change and not the inadequacy of the treatment. The ability of a measure to detect change is a particularly important consideration when the investigator is interested in measuring the improvements obtained by individual patients, as in the RC index approach.

Despite the difficulties with the RC approach noted above, it is important to recognize that the intent of this index is to overcome a common criticism with statistical procedures that are frequently used to test for outcome differences between treatment groups. Specifically, these procedures test for mean differences between groups and, therefore, average the magnitude of change across all patients within each treatment group. For example, although meta-analytic reviews (e.g., Blanchard et al. 1980; Holroyd and Penzien 1986) have generally supported the utility of relaxation and biofeedback for reduction of chronic headaches, it also has been found that 1/3 to 1/2 of headache sufferers fail to show what many would consider to be clinically significant benefits. Traditional statistical procedures designed to test for significant differences among group means do not test for the variability of treatment effects, although, of course, response variability can adversely affect the conclusions derived from these analyses. Without directly quantifying variability, there is no way of determining the proportion of patients who benefitted from treatment. These proportions, which can be calculated for example by computing an RC index for each patient, may be of particular importance to clinicians and third-party payers interested in estimating the probability that an individual patient will benefit from a specific therapeutic intervention.

Efficiency in the manner of administering treatment and cost-related criteria

Assuming that different techniques are equally effective on a particular outcome measure, the rapidity with these results are achieved becomes an important consideration. If a 3-week inpatient program produces the same results of an 8-week outpatient program, the inpatient program might be preferred. From a financial standpoint, however, the longer outpatient program might be less expensive and thus might be viewed as more cost-effective by third-party payers, even though they might need to make disability payments for an additional month. For example, in our own clinic at the University of Pittsburgh we have an 3.5-week all-day intensive rehabilitation program that costs approximately $9000. This can be compared with another program that we offer that consists of 8 weekly 3-h outpatient sessions. This 8-week extended program cost less than \$3000. Even with the additional month of disability payments (for worker compensation cases in the state of Pennsylvania minimum disability payments are \$133/week), the extended treatment program is still more cost-effective. From the patients' perspective, weekly outpatient sessions might be less disruptive and more acceptable than a 3-week hospital stay (Sturgis et al. 1984). Of course this assumes that both programs are equally effective. What data are there on the comparative effectiveness of inpatient and outpatient programs?

To our knowledge, only six studies, with varying degrees of methodological appropriateness (e.g., random assignment of patients to treatment conditions) have compared directly the efficacy of inpatient and outpatient treatments, with the results leading to totally opposite conclusions. Cairns et al. (1982) and Jarvikoski et al. (1986) reported initial advantages to inpatient treatment although differences between group dissipated over a 1-year follow-up. Cicala and Wright (1989) and Peters and Large (1990) reported essentially negligible differences between inpatient and outpatient programs; whereas Tait et al. (1988) reported a complex set of results with both impatient and outpatient groups improving but on different outcome measures. Harkapaa et al. (1989) reported superior 3-month follow-up results for an inpatient treatment. All of these studies suffered from methodological problems, the most serious and frequent being assignment of more dysfunctional cases into the inpatient treatment and thus biasing results against inpatient treatment.

Another way to consider efficacy of treatment is to consider the use of patient groups versus individual treatment. Spence (1989) demonstrated the efficacy of both group and individual cognitive-behavioral treatment for patients with occupational pain of the upper limbs. She found that on the majority of post-treatment and 6-month follow-up outcome measures the group and individual treatment programs were equally effective. However, the patients receiving the group treatment did reveal some deterioration in depression and anxiety measures at the time of follow-up. Moreover, the patients receiving the group treatment indicated that they were less satisfied with the treatment. Thus, there are apparently some advantages for the more costly individual treatment approach, although these results must be interpreted with some caution since the credibility of the two treatment approaches were not assessed and the treatments may not have been viewed by patients as equally credible. If these results are replicated they would raise the question as to whether attention should be devoted to strategies to enhance maintenance for the group treatment (Turk and Rudy 1991). Nonetheless, the results of this study underscore the importance of attending to consumer acceptance of the intervention provided as a relevant criterion of treatment outcome.

The setting in which treatment is provided also can be used to evaluate treatment efficacy and cost-effectiveness issues. For example, Attanasio et al. (1987) demonstrated that office-based versus largely home-based treatment for tension headache patients both resulted in clinically meaningful changes in headache activity (defined at 50% reduction of symptoms). Specifically, 71.4% of the patients in a cognitive-behavioral office-based treatment compared to 62.2% cognitive-behavioral home-based treatment reported changes in headaches. These results were accomplished with a 50–65% reduction in the amount of actual therapist time, and importantly, with relatively few drop-outs and a high rate of patient adherence to the treatment protocol for the home-based treatment. The results of Attanasio et al. are comparable to other home-based treatments for headache (e.g., Blanchard et al. 1985; Burke and Andrasik 1989; Richardson and McGrath 1989).

Corey et al. (1987) demonstrated the merits of a program of long-term home-based versus shorter term clinic-based programs. The home-based treatment program was 6 months in duration but, since it made use of paraprofessionals, it was less expensive than many clinic-based pain treatment programs. However, because the treatment effects required longer to achieve, insurance carriers had to continue paying disability for a period of time longer than traditional pain rehabilitation programs. Thus, this extended home-based program suggests a trade off among treatment expenses, disability payments, and the costs of keeping a case open.

Even if minimal professional therapist contact is not as effective as other treatment approaches, it may be more cost-effective and easier to disseminate. Additionally, this approach permits more patients to be treated and, therefore, may have a greater impact over all (Lorig et al. 1985; Lorig et al. 1986). To evaluate the overall impact, however, the question of how to weigh the cost-effectiveness of minimal contact against the apparent advantages of more intensive treatment contact needs to be considered.

The availability of professionals with the expertise to treat chronic pain is an additional concern. That is, the supply of trained professionals may not be adequate to meet the demand for pain treatment services. One approach to extending the supply of treatment personnel is to have professional staff supervise 'lay-helpers'. For example, Goeppinger et al. (1989) and Cohen et al. (1986) demonstrated the beneficial effects of self-care treatments for arthritics that made use of lay-helpers supervised by professional staff and 'home-study' programs. Such programs need to be compared to more costly programs that use greater amounts of professional contact. Similarly, even when professional involvement is demonstrated to be essential for success, the nature (group vs. individual) and extent of the contact (weekly vs. monthly) should be evaluated to determine the most efficient method of service delivery.

Additionally, non-professional self-help organizations such as the American Chronic Pain Association (with over 600 chapters in the United States, Canada, and Australia) may provide a valuable and cost-effective way to help maintain gains following successful treatment at a pain clinic. Home-based and community-based programs may be particularly valuable to foster maintenance and generalizations, given the high relapse rates reported in the second paper in this series (Turk and Rudy 1991), as they are embedded in the patient's natural environment.

Evaluating cost-effectiveness. There has recently appeared in the pain literature a number of studies that demonstrate the potential to teach family members, teachers, school nurses, community lay helpers, and formerly treated patients to serve as therapists, under professional guidance and supervision. These studies have demonstrated the cost-effectiveness of using lay therapists to treat both adults and children with chronic headaches, (e.g., Blanchard et al. 1985; Holroyd et al. 1988; Burke and Andrasik 1989), arthritis (Lorig et al. 1986; Goeppinger et al. 1989), and heterogeneous chronic pain sufferers (Cory et al. 1987).

One strategy to evaluate cost effectiveness quantitatively has been proposed by Burke and Andrasik (1989). To compare the relative cost-effectiveness of clinic-based versus home-based treatment for pediatric migraines, they divided the percentage of improvement in a headache composite index (derived from weekly diaries of frequency, duration, and intensity of headaches) from pre-treatment to follow-up by the amount of therapist time spent with the patients (3 h for home-based vs. 10 h for clinic-based treatment). Before contract time was considered, the two treatment approaches were roughly equivalent in significantly reducing headache indices. However, when the indices were adjusted for contract time the home-based treatment was found to be 3 times more cost efficient than the clinic-based treatment approach. A similar approach could be used to examine the comparative cost efficacy of any two treatment interventions (e.g., biofeedback vs. relaxation) or format (group vs. individual, inpatient vs. outpatient).

Lorig et al. (1985, 1986) demonstrated that lay-person arthritis self-management courses were as effective as professionally taught courses. Lorig et al. (1986) reported that the lay instructors were paid from \$0–\$100 for their time, in contrast to professionals who were paid from \$240–\$600 (\$20–60/h for 12 h) for their time teaching a course. Cost differences between these two treatment approaches can be best appreciated by considering the patient population for whom these services may be needed. In 1981, the American Arthritis Foundation estimated that there were over 30 million people in the United States who suffered from arthritis and related musculoskeletal diseases. Lorig et al. (1986) suggest that perhaps 1/3 of these individuals might benefit from health education and that if only 10% of these persons would attend groups with 20 persons/group, 50,000 groups would be required. Based on these figures, we can estimate the cost for lay versus professionally led groups. To conduct 50,000 groups would cost a maximum of \$5,000,000 for groups led by lay personnel, compared to \$30,000,000 for professionally led groups.

An intervention that can be widely disseminated, even if it is only moderately effective, may have greater impact on patient care than a more effective treatment approach that is more restricted in terms of the number of patients that can be treated. For example, although comprehensive, interdisciplinary pain clinics might be most desirable for treating patients with persistent pain, the use of individual pain specialists in rural communities or outreach programs using paraprofessionals (e.g., Corey et al. 1987) may be the best that can be offered.

Provided acceptable outcome criteria demonstrate that several treatment approaches produce beneficial effects, the therapeutic intervention that can be disseminated more widely should be preferred since it is also the treatment approach that is likely to be the more cost-effective. Consider the potential audience that can be reached with a 'self-help' book such as those developed by Lorig and Fries (1980), Catalano (1987), Corey (1988), and McGrath et al. (1990), compared to the potential population of chronic pain sufferers who can be treated in pain clinics.

If the use of group interventions, home self-care, and non-professionals can be demonstrated to have a beneficial effect, the burden of proof for professionals will be to demonstrate that the additional cost of their individual and or inpatient services is merited, both from a treatment efficacy and cost-effective standpoint. Some readers might be concerned that we have given too much attention to financial issues in our consideration of treatment efficacy. As treatment costs rise and third party reimbursements decrease, providers will increasingly be called upon to produce maximal benefits at minimal costs. Therefore, it becomes prudent to isolate shared components of effective therapies and determine the best way to provide them for cost-effective outcomes (Simmons et al. 1988; Stieg and Turk 1988; Tobin et al. 1988). This is not meant to suggest that health care providers should abrogate their responsibility to recommend and provide what they believe is the most appropriate treatment based on clinical needs. However, with the large number of third-party payers denying payment (see Turk and Rudy 1990) it behooves health care providers to at least acknowledge and be responsive to the limited resources of third-party payers (Stieg and Turk 1988).

Conclusion

Hindsight is an exact science and as such is always much more accurate than foresight. Moreover, it is a truism that we tend to be more critical of other people's research than our own. We are quite willing to offer criticisms but generally tend to be less adventurous when it comes to the actual execution of research. Based on the wide range of issues that we have raised in this third paper and the two preceding papers in this series (Turk and Rudy 1990, 1991), it is quite obvious that conducting treatment outcome research with all of its problems suggests why so few pain treatment outcome studies actually appear in the literature. Undertaking treatment outcome research is a daunting task indeed. We should be humbled by the complexity of treatment outcome research, but hopefully not deterred from systematically evaluating the effectiveness of the treatments that we provide to individuals suffering from chronic pain.

The 'perfect' treatment outcome study is not only inconceivable, but it would surely be impossible to execute. Nonetheless, we need to continue to refine our measures, and improve upon our experimental designs and statistical methods. This includes broadening our definition of what constitutes 'good' pain treatment outcome research methods to incorporate well designed studies that make use of comparative and quasi-experimental research methods. That is, in many areas of pain rehabilitation research it may be unethical and virtually impossible to design placebo control groups in order to compare the effectiveness of an experimental treatment condition. The illusion that placebo control conditions constitute good research has been addressed in the psychotherapy research literature (e.g., Parloff 1986), and has led some psychotherapy research methodologists to abandon this research model in favor of comparative outcome studies (e.g., Kazdin 1986). These issues as they relate to pain treatment outcome studies have recently been addressed cogently by Peters and Large (1990).

We also need to develop some consensus about what criteria of success should be used to measure treatments for chronic pain and how these constructs should be operationalized. We need to consider some of the alternative criteria for determining success described throughout this paper rather than continuing to rely exclusively on traditional mean group treatment outcome on self-report measures. Finally, we need to be cautious in what we claim about our treatment approaches and we must acknowledge the strengths and limitations of our studies.

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